

PAPER_1418 — UQFF: Cosmic Ray Ankle (Bucket H)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket H **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — formal catalog tuple in Bucket H **Calculator surface:** `calculate_high_energy_astro({...})`

Closure

$E_{\text{ankle}} = m_p \times D_{\text{crit}}^7 / K_{\text{MEX}} = 3.62e18 \text{ eV}$ (20% vs Auger $3e18$)

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_high_energy_astro_report()` or equivalent surface in `uqff_pure_calculator.py`.
 - Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.
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